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Paper Code : PEC-CS701E/PEC-IT701D/PECICB702A Machine learning

UPID : 007554

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) Fraud Detection, Image Classification, Diagnostic, and Customer Retention are applications in _____ learning algorithm.
- (II) Machine learning algorithm build a model which is based on sample data is known as _____.
- (III) The prior goal of unsupervised learning model is to determine the _____.
- (IV) Real-Time decisions, Game AI, Learning Tasks, Skill Aquisition, and Robot Navigation are applications in _____.
- (V) Deep learning algorithms are _____ more accurate than machine learning algorithm in image classification.
- (VI) True or False: Overfitting is more likely when you have huge amount of data to train.
- (VII) If you use an ensemble of different base models, is it necessary to tune the hyper parameters of all base models to improve the ensemble performance?
- (VIII) The Bayes rule can be used in answering _____.
- (IX) True or False: K-means clustering aims to partition n observations into k clusters.
- (X) Which ensemble model helps in reducing variance?
- (XI) Write one common feature about neural network and linear regression models.
- (XII) In a Decision Tree Leaf Node represents _____.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. How to Tackle Overfitting and Underfitting? [5]
3. Explain Pool-Based sampling in Active learning. [5]
4. Differentiate Between Machine Learning and Deep Learning. [5]
5. Explain How a system can play a Game of Chess using Reinforcement Learning. [5]
6. Explain 'Naive' in a Naive Bayes Algorithm. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) What Is a False Positive and False Negative and How are they significant with an example? [5]
- (b) Why do you need Confusion matrix? [4]
- (c) For the given dataset, apply Naive Baye's Algorithm and predict the outcome for a car= {Red, Domestic, SUV}. [6]

$$X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Determine the $x(n)$ using time –shifting property.

8. (a) Explain: Sparse Modelling and estimation. [5]
- (b) Explain: Time series data in machine learning [5]
- (c) Explain how deep learning and feature representation learning related. [5]
9. (a) Explain the ID3 algorithm for decision tree learning. [4]
- (b) What is the procedure of building Decision tree using ID3 with Gain and Entropy. [5]
- (c) What do you mean by Information gain and Entropy and its relation with ID3. [6]
10. (a) Why KNN is non-parametric? [3]

- (b) Write down three Pros and Cons of KNN algorithm. [6]
- (c) What is Euclidean distance in terms of machine learning? Using Euclidean distance find the distance between points P(3, 2) and Q(4, 1). [6]
11. (a) What is model selection? Write down the types of data and based on that which model is used? [5]
- (b) Explain Cross validation with an example. [5]
- (c) Explain Statistical Learning theory. [5]

*** END OF PAPER ***